

Over the past 75 years the study of kaon physics has been integral to the development of the standard model. Since 2016, the NA62 experiment has been the leading experiment in the field of kaon physics. However, the future of kaon physics is uncertain, with the NA62 experiment funded up to CERN LS3, which is expected at the end of 2025. It was then required to investigate the future of kaon physics and the potential experiments that could succeed NA62. This chapter will discuss the potential future of kaon physics, the experiments that could be used to further the field, and the work contributed to these experiments.

0.1 High Intensity Kaon Experiments (HIKE) Description

The High-intensity kaon experiments (HIKE) was a proposed project to continue kaon physics at CERN in the ECN3 experimental hall and represented a long-phased plan after LS3. HIKE would have profited from a four to six-fold increase in beam intensity compared to NA62 and included cutting-edge detector technologies that will be described in this section. There were three phases within the HIKE program with the first being a continuation of NA62's charged kaon physics with the high-intensity beam and detector upgrades. The second would have involved neutral kaon physics with charged secondary particle tracking, and the third phase would have been designed to measure the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay.

0.1.1 Phase 1

The experimental design of NA62 has been proven as a successful setup for the precision measurement of the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay. The HIKE phase 1 detector setup is shown in figure 1 with new and upgraded detectors with respect to the NA62 setup described in Section ???. The new and upgraded detectors are required due to the four times greater particle rate with respect to NA62 equating to 1.2×10^{13} protons per spill on the target. Other than the intensity upgrade, the beam had similar characteristics to NA62 with a $75 \text{ GeV}/c$ secondary beam produced from the $400 \text{ GeV}/c$ primary beam from the SPS accelerator onto a target at an angle of 0 rad .

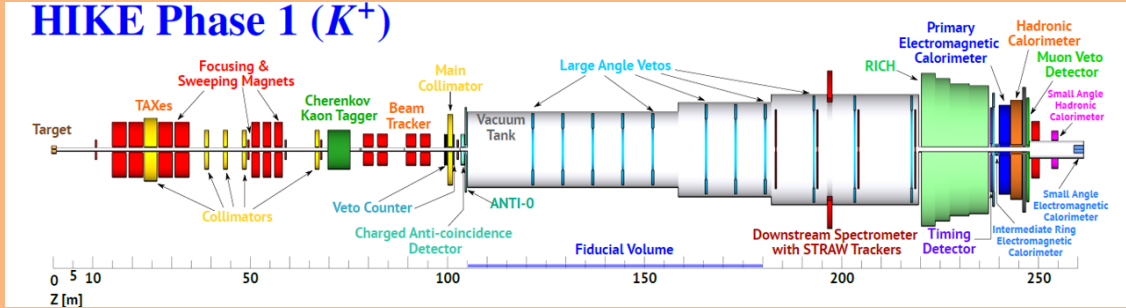


Figure 1: Diagram of HIKE phase 1 detector setup [?]

To sustain the increased kaon rate, the KTAG required upgrades to the photon-detection system and data acquisition system, with an expected time resolution of 15 - 20 ps and a kaon identification efficiency of greater than 95 %. The expected kaon rate was 200 MHz which corresponds to a maximum particle rate $10 \text{ MHz}/\text{cm}^2$ of photons at the PMT arrays, taking into the production of Cherenkov photons in CEDAR-H and the resultant acceptance of the photons. It was proposed that the PMT arrays would be replaced with arrays of micro-channel plate photomultipliers (MCP-PMTs), which provide a single photon time resolution of 50 - 70 ps depending

on the model, a low dark rate of 1 kHz/cm^2 and the ability to sustain the required photon rate. The downside of the MCP-PMTs is their limited lifetime from the aging of the Photo-Cathode caused by heavy feedback ions from the residual gas. This results in the rapid degradation in the PMT quantum efficiency with the increasing integrated anode charge as it is used.

0.1.2 Phase 2

0.1.3 Phase 3

0.2 RICH upgrade design

0.3 GTk upgrade design

0.4 Detector Rates